Longest Continuous Subarray (medium)

Problem Statement

Find the length of the longest contiguous subarray within an array of integers, where the absolute difference between any two elements in this subarray does not exceed a specified limit. The challenge lies in ensuring the subarray is continuous and the range (difference between the maximum and minimum element in the subarray) fits within the given threshold.

Examples

1. **Example 1**:
   - **Input**: Array = [10, 1, 2, 4, 7], Limit = 5
   - **Expected Output**: 3
   - **Justification**: The longest subarray where the absolute difference between any two numbers is at most 5 is [1, 2, 4].
2. **Example 2**:
   - **Input**: Array = [4, 8, 5, 1, 7, 9], Limit = 3
   - **Expected Output**: 2
   - **Justification**: One possible longest subarray is [7, 9] where the difference between 7 and 9 is within the limit of 3.
3. **Example 3**:
   - **Input**: Array = [3, 3, 3, 3, 3], Limit = 0
   - **Expected Output**: 5
   - **Justification**: Since all elements are the same, the entire array is the longest subarray with differences of 0.

Try it yourself

Try solving this question here:

Java

```java
import java.util.TreeMap;

public class Solution {
    // Function to find the longest subarray with absolute diff less than or equal to li
    public  int longestSubarray(int[] nums, int limit) {
        int left = 0, maxLength = 0;
        // ToDo: Write Your Code Here.
        return maxLength;
    }
}

```

Test Case | Run Code Result | Stdout

```
[8,2,4,7]
4
```

Use Example Testcases | Run | Submit